

Beyond one shot: agentic LaTeX → Typst conversion

Four one-factor upgrades, ablated on the 6 hardest lathe benchmark samples

37 → 82

mean overall fidelity, 1-shot baseline vs
best method (core 6)

\$69

total tracked API spend

68

runs completed

Setup - task, metric, and gates

Task set

The 6 hardest of 157 benchmark samples (one-shot LLM overall 31-38/100): four table documents that reflow to the wrong page count, one algorithm block, one figure+caption page. Reference = pdfLaTeX render.

Pass gates (frozen before running)

G1 compiles - G2 page count and pagination perfect - G3 token precision AND recall ≥ 0.95 - G4 layout ≥ 75 .

Score (deterministic, 0-100)

$\text{overall} = \text{content}^{0.35} \times \text{visual}^{0.65}$; $\text{visual} = \text{pagination}^{0.10} \times \text{layout}^{0.40} \times \text{typography}^{0.20} \times \text{raster}^{0.30}$. Raster uses the raster_v0.2 rescoring (tolerant ink F1 + edge distance) which removes v0.1's ~25-30 harshness ceiling; identity still scores 100 and orderings are preserved.

How to read this deck

Each following section changes exactly ONE factor (prompt->harness, effort, visual feedback, tooling) and shows the same table: per-metric means over the core 6, cost, duration, iterations - then a case study with the agent's own reasoning per scoring iteration.

Leaderboard - core 6 hardest samples

all metrics 0-100; mean over samples

model	effort	feedback	harness	overall	content	layout	typo	raster	pagin.	pass	\$/run
opus-4.7	medium	visual	v3	81.6	95.0	80.6	83.5	58.9	100.0	2/6	1.18
opus-4.7	medium	visual	v2	78.2	96.4	77.2	83.7	51.7	99.1	3/6	1.30
opus-4.7	high	blind	v1	(76.6)	(96.9)	(82.8)	(87.7)	(42.8)	(100.0)	2/4	2.45
opus-4.7	high	visual	v1	(76.0)	(96.9)	(77.4)	(83.6)	(42.3)	(100.0)	3/5	2.65
opus-4.7	medium	visual	v1	75.1	96.6	71.4	81.5	46.8	99.7	2/6	1.53
sonnet-4.6	low	visual	v1	70.6	95.6	67.7	78.7	38.0	100.0	0/6	1.84
opus-4.7	low	visual	v1	(69.9)	(95.3)	(71.7)	(77.8)	(33.5)	(100.0)	1/2	-
sonnet-4.6	low	blind	v1	(67.5)	(93.8)	(68.9)	(84.5)	(30.2)	(100.0)	0/5	0.99
opus-4.7	medium	blind	v1	67.4	94.7	70.5	81.5	28.6	100.0	0/6	0.88
opus-4.7	low	blind	v1	66.8	94.1	65.0	84.6	32.7	92.5	2/6	-
sonnet-4.6	low	1-turn	—	(43.4)	(90.1)	(34.4)	(76.2)	(10.4)	(65.4)	0/6	0.10
opus-4.7	low	1-turn	—	(41.5)	(89.4)	(34.0)	(71.4)	(9.6)	(60.0)	0/6	-
gemini-3.1-flash-lite	—	1-turn	—	37.0	83.5	30.6	76.1	7.3	56.9	0/6	-

Parenthesized = partial coverage (queues still running). The four highlighted one-factor steps from the bottom of this table to the top are ablated in the following sections.

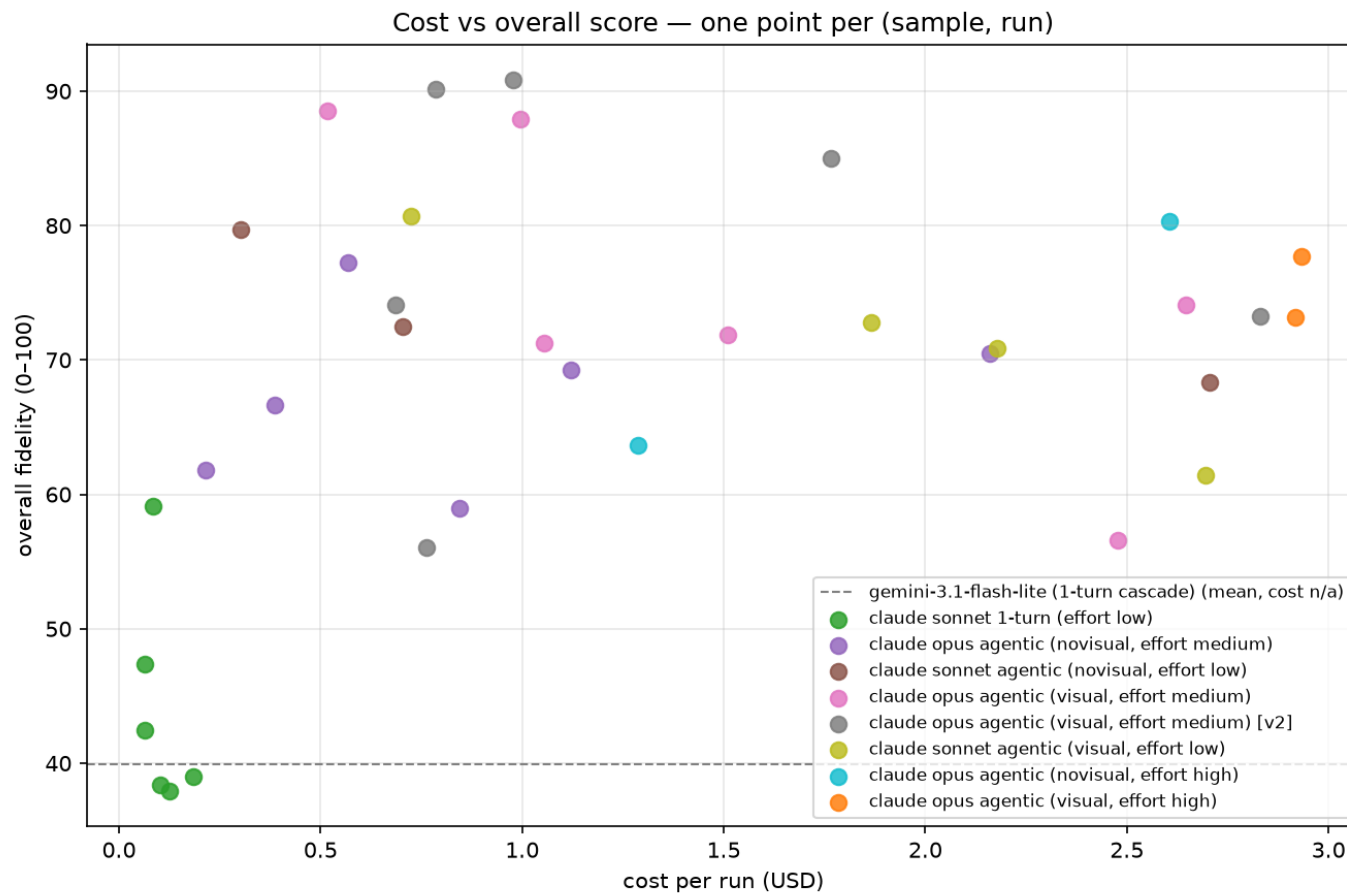
Reading the plot:

- One-shot points (bottom-left, dashed gemini line) cluster at 32-52 regardless of model.

- Best value sits at \$0.5-1.0 per document.

- Spending to the \$3 cap does not buy score: those points are budget-cap grinders oscillating around a plateau.

- Implication: feedback loop >> model scale >> raw budget.



Upgrade 1: one-shot prompt -> agentic harness

What changed in this step

Same model (opus-4.7), same low effort, still no images. The only change: instead of one prompt -> one answer, the model gets a workdir with main.tex, .score.sh (compile + official metrics), and freedom to edit-score-edit under a \$3 cap. Feedback is numeric only (blind).

	overall	content	layout	typo	raster	pagin.	pass	\$/run	min/run	iters
BEFORE opus-4.7 / low / 1-turn	(41.5)	(89.4)	(34.0)	(71.4)	(9.6)	(60.0)	0/6	-	-	-
AFTER opus-4.7 / low / blind / v1	66.8	94.1	65.0	84.6	32.7	92.5	2/6	-	-	-
Δ	+25.3	+4.7	+31.0	+13.2	+23.1	+32.5	+2	-	-	-

Per-sample overall (before -> after):

06_tables_moderate_010: 39.0 -> 76.9 (+38.0)
05_tables_simple_023: 35.4 -> 68.4 (+33.0)
05_tables_simple_005: 38.0 -> 82.9 (+44.9)
09_algorithms_003: 56.5 -> 46.6 (-9.9)
05_tables_simple_021: 38.6 -> 43.9 (+5.3)
07_figures_captions_007: nc/- -> 81.9

	overall	content	layout	typo	raster	pagin.
BEFORE opus-4.7 / low / 1-turn	38.0	86.9	24.6	65.1	9.6	53.5
AFTER opus-4.7 / low / blind / v1	82.9	99.9	78.5	87.5	57.8	100.0

No transcript for the AFTER run (older run without step logging).

Upgrade 2: reasoning effort low -> medium

What changed in this step

Identical harness, identical blind feedback. Only the CLI --effort flag changes, giving the model a larger thinking budget per turn.

	overall	content	layout	typo	raster	pagin.	pass	\$/run	min/run	iters
BEFORE opus-4.7 / low / blind / v1	66.8	94.1	65.0	84.6	32.7	92.5	2/6	-	-	-
AFTER opus-4.7 / medium / blind / v1	67.4	94.7	70.5	81.5	28.6	100.0	0/6	0.88	2.8	8
Δ	+0.6	+0.6	+5.5	-3.2	-4.1	+7.5	-2	-	-	-

Per-sample overall (before -> after):

- 06_tables_moderate_010: 76.9 -> 69.3 (-7.6)
- 05_tables_simple_023: 68.4 -> 70.5 (+2.1)
- 05_tables_simple_005: 82.9 -> 77.2 (-5.6)
- 09_algorithms_003: 46.6 -> 61.8 (+15.2)
- 05_tables_simple_021: 43.9 -> 59.0 (+15.0)
- 07_figures_captions_007: 81.9 -> 66.6 (-15.3)

	overall	content	layout	typo	raster	pagin.
BEFORE opus-4.7 / low / blind / v1	46.6	94.1	36.7	85.3	10.1	79.9
AFTER opus-4.7 / medium / blind / v1	61.8	94.9	71.2	80.8	16.9	100.0

AFTER run, iteration by iteration (opus-4.7 / medium / blind / v1):

iter	overall	Δ	agent's reasoning before this score run / last edit
#01	55.1	start	write: "#set page(paper: "us-letter", margin: 1in) #set text(size: 11pt, font: "New Comp"

Upgrade 3: blind -> visual feedback

What changed in this step

Same model, effort, harness. The agent may now Read reference page PNGs and the labeled diff images (red = missing, blue = extra, yellow/orange = displaced) regenerated by every score run.

	overall	content	layout	typo	raster	pagin.	pass	\$/run	min/run	iters
BEFORE opus-4.7 / medium / blind / v1	67.4	94.7	70.5	81.5	28.6	100.0	0/6	0.88	2.8	8
AFTER opus-4.7 / medium / visual / v1	75.1	96.6	71.4	81.5	46.8	99.7	2/6	1.53	4.4	10
Δ	+7.6	+1.9	+1.0	-0.0	+18.2	-0.3	+2	+0.65	-	-

Per-sample overall (before -> after):

- 06_tables_moderate_010: 69.3 -> 74.1 (+4.8)
- 05_tables_simple_023: 70.5 -> 56.6 (-13.9)
- 05_tables_simple_005: 77.2 -> 71.3 (-6.0)
- 09_algorithms_003: 61.8 -> 87.9 (+26.1)
- 05_tables_simple_021: 59.0 -> 71.9 (+12.9)
- 07_figures_captions_007: 66.6 -> 88.5 (+21.9)

	overall	content	layout	typo	raster	pagin.
BEFORE opus-4.7 / medium / blind / v1	61.8	94.9	71.2	80.8	16.9	100.0
AFTER opus-4.7 / medium / visual / v1	87.9	96.8	92.0	85.4	67.9	100.0

AFTER run, iteration by iteration (opus-4.7 / medium / visual / v1):

iter	overall	Δ	agent's reasoning before this score run / last edit
#01	61.2	start	edit: "#ind(2)[*if* \$P_(u,t) > #text(font: "DejaVu Sans Mono", size: 10pt)[MinimumPower"
#02	61.2	+0.0	edit: "#set par(justify: true, leading: 0.55em, first-line-indent: 0pt)"
#03	72.9	+11.7	edit: "#set par(justify: false, leading: 0.45em, spacing: 0.5em) #show math.equation: s"
#04	74.0	+1.1	edit: "#set par(justify: false, leading: 0.5em, spacing: 0.55em) #show math.equation: s"
#05	78.2	+4.2	edit: "#set par(justify: false, leading: 0.55em, spacing: 0.7em) #show math.equation: s"
#06	74.6	-3.6	edit: "#v(-0.5cm) #line(length: 100%, stroke: 0.8pt) #v(-0.5em) #line(length: 100%, str"
#07	78.2	+3.6	edit: "#line(length: 100%, stroke: 0.8pt) #v(-0.4em) #line(length: 100%, stroke: 0.4pt)"

Upgrade 4: harness v1 -> v2 tooling

What changed in this step

Same model, effort, visual feedback. v2 adds: best-so-far auto-checkpoint (regressions become free); ./sweep.sh deterministic grid search over typst --input values; fine-grained per-component AND per-page score report; registered-raster diagnostic (separates misalignment from the font floor).

	overall	content	layout	typo	raster	pagin.	pass	\$/run	min/run	iters
BEFORE opus-4.7 / medium / visual / v1	75.1	96.6	71.4	81.5	46.8	99.7	2/6	1.53	4.4	10
AFTER opus-4.7 / medium / visual / v2	78.2	96.4	77.2	83.7	51.7	99.1	3/6	1.30	4.2	8
Δ	+3.2	-0.2	+5.8	+2.2	+4.9	-0.5	+1	-0.23	-	-

Per-sample overall (before -> after):

- 06_tables_moderate_010: 74.1 -> 74.1 (-0.0)
- 05_tables_simple_023: 56.6 -> 73.3 (+16.7)
- 05_tables_simple_005: 71.3 -> 90.8 (+19.6)
- 09_algorithms_003: 87.9 -> 85.0 (-2.9)
- 05_tables_simple_021: 71.9 -> 56.1 (-15.8)
- 07_figures_captions_007: 88.5 -> 90.1 (+1.6)

	overall	content	layout	typo	raster	pagin.
BEFORE opus-4.7 / medium / visual / v1	71.3	99.9	58.2	90.1	38.9	100.0
AFTER opus-4.7 / medium / visual / v2	90.8	99.4	87.7	89.9	78.9	100.0

AFTER run, iteration by iteration (opus-4.7 / medium / visual / v2):

iter	overall	Δ	agent's reasoning before this score run / last edit
#01	68.4	start	write: "#set page(paper: "us-letter", margin: 1in) #set text(font: "New Computer Modern""
#02	65.5	-2.9	Content is 99. Table rows are taller than reference and shifted down. Let me tighten row spacing. edit: " inset: (x: 4pt, y: 1.2pt), table.header(text(size: 8pt)[*Index*], text(size: 8"
#03	68.6	+3.1	edit: " inset: (x: 4pt, y: 1.5pt), table.header(text(size: 7pt)[*Index*], text(size: 7"
#04	68.6	+0.0	edit: "#let topmargin = eval(sys.inputs.at("topmargin", default: "1in")) #set page(pape"
#05	76.4	+7.8	0 edit(s) before scoring
#06	76.4	+0.0	0 edit(s) before scoring
#07	80.7	+4.3	edit: "#pagebreak() #v(p2gap) #make_table[Table 2: Source table 2: 2512.00323_table_10]"
#08	82.0	+1.3	0 edit(s) before scoring
#09	82.0	+0.0	edit: "#let p2gap = eval(sys.inputs.at("p2gap", default: "70pt"))"

Upgrade 5: harness v2 -> v3 (fix the mid-run reward)

What changed in this step

Same model, effort, visual feedback, and v2 tooling. One change: the score the agent sees MID-RUN now uses raster_v0.2 (tolerant ink F1 + edge distance) - the same number as final grading - instead of v0.1's harsh raster whose ~25-30 ceiling made real improvements invisible mid-run. The agent stops wasting budget chasing a dead component and optimizes what actually moves.

	overall	content	layout	typo	raster	pagin.	pass	\$/run	min/run	iters
BEFORE opus-4.7 / medium / visual / v2	78.2	96.4	77.2	83.7	51.7	99.1	3/6	1.30	4.2	8
AFTER opus-4.7 / medium / visual / v3	81.6	95.0	80.6	83.5	58.9	100.0	2/6	1.18	3.6	9
Δ	+3.3	-1.5	+3.4	-0.1	+7.2	+0.9	-1	-0.12	-	-

Per-sample overall (before -> after):

- 06_tables_moderate_010: 74.1 -> 86.4 (+12.3)
- 05_tables_simple_023: 73.3 -> 64.4 (-8.9)
- 05_tables_simple_005: 90.8 -> 88.0 (-2.8)
- 09_algorithms_003: 85.0 -> 86.8 (+1.7)
- 05_tables_simple_021: 56.1 -> 74.8 (+18.7)
- 07_figures_captions_007: 90.1 -> 89.2 (-0.9)

	overall	content	layout	typo	raster	pagin.
BEFORE opus-4.7 / medium / visual / v2	56.1	90.4	42.9	65.5	25.7	94.7
AFTER opus-4.7 / medium / visual / v3	74.8	88.8	71.2	77.5	52.0	99.9

AFTER run, iteration by iteration (opus-4.7 / medium / visual / v3):

iter	overall	Δ	agent's reasoning before this score run / last edit
#01	53.8	start	write: "#set page(paper: "us-letter", margin: 1in) #set text(font: "New Computer Modern""
#02	61.0	+7.2	Table 2 needs to be more compact. Let me reduce font size and row spacing. edit: "// ===== TABLE 2 ===== #set text(size: 8pt) #set par(leading: 0.4e"
#03	63.1	+2.1	edit: "// ===== TABLE 2 ===== #set text(size: 8pt) #set par(leading: 0.4e"

Reference (pdfLaTeX)

Simple Tables

Source-backed grouped table sample

1 Tables

These tables are grouped from source-backed LaTeX table data and compiled as a single benchmark data point.

Table 1: Source table 1: 2512.00323_table_8

Index	Model Name	Model Size (MB)	Dataset
1	facebook/abert-base-v2-squad2	12	SQuAD v2
2	valhalla/bert-large-finetuned-squadv1	388	SQuAD v1
3	dropout/bert-base-cased-squad2	354	SQuAD v2
4	google/highlight-roberta-base	122	Book, CC-News, Stories and Wikipedia
5	google/highlight-roberta-base	122	Book, CC-News, Stories and Wikipedia
6	dmu-uh/bert-v1	104	NA
7	dropout/roberta-base-squad2	119	SQuAD v2
8	OpenAI/gpt3-medium	136	NA
9	YituTech/ernn-bert-base	101	NA
10	Public-muon/bert-base-squad	307	SQuAD v1
11	muonmuon/abert-v2-large	814	NA
12	distilbert-base-microsoft	64	BookCorpus and English Wikipedia
13	hlsdhrd/ernn-electra-base-squad2	104	SQuAD v2
14	mlcommons/ernn-1.0-base-ah	96	Chinese
15	shu-nlu-en-2018	637	Machine language modeling
16	google/bert-base	80	Common Crawl, CoNLL-X, and Common Crawl
17	facebook/transformer-small	125	BookCorpus, English Wikipedia, CoNLL-X, CoNLL-X, and Common Crawl
18	distilbert-base-v1	125	CoNLL-X
19	hlsdhrd/ernn-1.0-base-ah	96	Chinese
20	gpt2	119	NA
21	ernn/bert-base	119	NA
22	abert-base-1024	115	NA
23	abert-base-1024	115	NA
24	facebook/abert-base-v2	115	NA
25	ernn/bert-base	115	NA
26	ernn/bert-base	115	NA
27	ernn/bert-base	115	NA
28	ernn/bert-base	115	NA
29	ernn/bert-base	115	NA
30	ernn/bert-base	115	NA
31	ernn/bert-base	115	NA
32	ernn/bert-base	115	NA
33	ernn/bert-base	115	NA
34	ernn/bert-base	115	NA
35	ernn/bert-base	115	NA
36	ernn/bert-base	115	NA
37	ernn/bert-base	115	NA
38	ernn/bert-base	115	NA
39	ernn/bert-base	115	NA
40	ernn/bert-base	115	NA
41	ernn/bert-base	115	NA
42	ernn/bert-base	115	NA
43	ernn/bert-base	115	NA
44	ernn/bert-base	115	NA
45	ernn/bert-base	115	NA
46	ernn/bert-base	115	NA
47	ernn/bert-base	115	NA

Final candidate (Typst)

Simple Tables

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2	valhalla/bert-large-finetuned-squadv1	388	SQuAD v1
3	dropout/bert-base-cased-squad2	354	SQuAD v2
4	google/highlight-roberta-base	122	Book, CC-News, Stories and Wikipedia
5	google/highlight-roberta-base	122	Book, CC-News, Stories and Wikipedia
6	dmu-uh/bert-v1	104	NA
7	dropout/roberta-base-squad2	119	SQuAD v2
8	OpenAI/gpt3-medium	136	NA
9	YituTech/ernn-bert-base	101	NA
10	Public-muon/bert-base-squad	307	SQuAD v1
11	muonmuon/abert-v2-large	814	NA
12	distilbert-base-microsoft	64	BookCorpus and English Wikipedia
13	hlsdhrd/ernn-electra-base-squad2	104	SQuAD v2
14	mlcommons/ernn-1.0-base-ah	96	Chinese
15	shu-nlu-en-2018	637	Machine language modeling
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17	facebook/transformer-small	125	BookCorpus, English Wikipedia, CoNLL-X, CoNLL-X, and Common Crawl
18	distilbert-base-v1	125	CoNLL-X
19	hlsdhrd/ernn-1.0-base-ah	96	Chinese
20	gpt2	119	NA
21	ernn/bert-base	119	NA
22	abert-base-1024	115	NA
23	abert-base-1024	115	NA
24	facebook/abert-base-v2	115	NA
25	ernn/bert-base	115	NA
26	ernn/bert-base	115	NA
27	ernn/bert-base	115	NA
28	ernn/bert-base	115	NA
29	ernn/bert-base	115	NA
30	ernn/bert-base	115	NA
31	ernn/bert-base	115	NA
32	ernn/bert-base	115	NA
33	ernn/bert-base	115	NA
34	ernn/bert-base	115	NA
35	ernn/bert-base	115	NA
36	ernn/bert-base	115	NA
37	ernn/bert-base	115	NA
38	ernn/bert-base	115	NA
39	ernn/bert-base	115	NA
40	ernn/bert-base	115	NA
41	ernn/bert-base	115	NA
42	ernn/bert-base	115	NA
43	ernn/bert-base	115	NA
44	ernn/bert-base	115	NA
45	ernn/bert-base	115	NA
46	ernn/bert-base	115	NA
47	ernn/bert-base	115	NA

Raster diff (red=missing, blue=extra)

Simple Tables

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3	dropout/bert-base-cased-squad2	354	SQuAD v2
4	google/highlight-roberta-base	122	Book, CC-News, Stories and Wikipedia
5	google/highlight-roberta-base	122	Book, CC-News, Stories and Wikipedia
6	dmu-uh/bert-v1	104	NA
7	dropout/roberta-base-squad2	119	SQuAD v2
8	OpenAI/gpt3-medium	136	NA
9	YituTech/ernn-bert-base	101	NA
10	Public-muon/bert-base-squad	307	SQuAD v1
11	muonmuon/abert-v2-large	814	NA
12	distilbert-base-microsoft	64	BookCorpus and English Wikipedia
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18	distilbert-base-v1	125	CoNLL-X
19	hlsdhrd/ernn-1.0-base-ah	96	Chinese
20	gpt2	119	NA
21	ernn/bert-base	119	NA
22	abert-base-1024	115	NA
23	abert-base-1024	115	NA
24	facebook/abert-base-v2	115	NA
25	ernn/bert-base	115	NA
26	ernn/bert-base	115	NA
27	ernn/bert-base	115	NA
28	ernn/bert-base	115	NA
29	ernn/bert-base	115	NA
30	ernn/bert-base	115	NA
31	ernn/bert-base	115	NA
32	ernn/bert-base	115	NA
33	ernn/bert-base	115	NA
34	ernn/bert-base	115	NA
35	ernn/bert-base	115	NA
36	ernn/bert-base	115	NA
37	ernn/bert-base	115	NA
38	ernn/bert-base	115	NA
39	ernn/bert-base	115	NA
40	ernn/bert-base	115	NA
41	ernn/bert-base	115	NA
42	ernn/bert-base	115	NA
43	ernn/bert-base	115	NA
44	ernn/bert-base	115	NA
45	ernn/bert-base	115	NA
46	ernn/bert-base	115	NA
47	ernn/bert-base	115	NA

Findings and next steps

Which upgrades pay

1-turn -> harness is the big one (+~20 overall; page counts fixed everywhere). Visual feedback and v2 tooling each add several points; extra effort mainly buys reliability, not peak score. Model scale alone changes almost nothing at 1 turn.

v2 tooling

Checkpointing makes bad experiments free; the deterministic sweep found margins/spacing the LLM had hand-nudged for 15 turns - runs that used it posted the two best scores of the experiment.

Where it saturates

Raster sits at its font floor (~20-30); late iterations oscillate by ~1 point. 05_tables_simple_021 resists all variants so far.

Next

v2.1 prompt: always sweep before manual nudges, keep iterating while under budget. Repeat runs for noise; human-calibrate metric weights; then RL hill-climbing with the same deterministic reward.